

EV SHOP GH

Electric Vehicle Parts & Sourcing Platform - Project Documentation

Executive Summary

Project Overview & Mission

EV Shop GH is Ghana's premier e-commerce platform dedicated to electric vehicle (EV) parts, accessories, and international vehicle sourcing services. The platform bridges the gap between global EV supply chains and the Ghanaian market, offering everything from lithium battery packs and charging stations to complete motorcycle and tricycle conversion kits.

The project was architected as a full-stack Node.js application with a refined, Porsche-inspired user interface emphasizing minimalism, sharp typography, and cinematic product presentation. Every design decision prioritizes trust, clarity, and conversion optimization for the West African EV market.

PRODUCTS	CATEGORIES	TECH STACK	PAYMENTS
16 SKUs	6 Verticals	Node + SQLite	Paystack

Business Model & Revenue Streams

How EV Shop GH Generates Value

Primary Revenue Streams

Direct Product Sales

High-margin EV components sold via e-commerce checkout with Paystack integration. Products include LiFePO4 battery packs (GHS 850-42,000), charging stations (GHS 850-9,500), BLDC motors (GHS 1,800-7,500), and complete conversion kits (GHS 5,800-12,500).

Vehicle Sourcing Service

End-to-end international procurement service sourcing EVs from China, Europe, and North America. Includes customs clearance, shipping logistics, and port delivery to Tema, Takoradi, Kumasi, or custom destinations. Commission-based or flat-fee pricing model.

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Technical Consultation

WhatsApp-based EV engineering support for installation, compatibility queries, and system design. Builds long-term customer relationships and drives repeat parts purchases.

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Product Catalog

Complete Inventory & Pricing Structure

Battery Systems

High-voltage energy storage for motorcycles, cars, and commercial EVs

SKU	Product Name	Price	Original	Save	Stock
BAT-001	LiFePO4 Battery Pack 72V 100Ah	GHS 8,500	GHS 9,800	13%	12
BAT-002	BMS 48V 100A Smart Battery Management	GHS 1,200	GHS 1,500	20%	25
BAT-003	LiNMC Battery Module 400V 60kWh	GHS 42,000	-	-	4

Charging Equipment

AC/DC charging infrastructure for residential and commercial deployment

SKU	Product Name	Price	Original	Save	Stock
CHG-001	7kW Home Wall Charger (Type 2)	GHS 3,200	GHS 3,800	16%	18
CHG-002	22kW AC Commercial Charger	GHS 9,500	GHS 11,000	14%	6
CHG-003	Portable 3.5kW Travel Charger	GHS 850	GHS 1,100	23%	35

Motors & Controllers

Powertrain components for EV conversions and new builds

SKU	Product Name	Price	Original	Save	Stock
MOT-001	BLDC Hub Motor 3000W 72V	GHS 2,800	GHS 3,200	12%	20
MOT-002	AC Induction Motor 15kW 96V	GHS 7,500	-	-	8
MOT-003	Motor Controller 72V 300A MOSFET	GHS 1,800	GHS 2,200	18%	15

Cooling & Thermal

Thermal management for Ghanaian climate conditions

SKU	Product Name	Price	Original	Save	Stock
COOL-001	Battery Thermal Management Kit	GHS 3,500	GHS 4,000	12%	9
COOL-002	EV Motor Cooling Jacket 72-96V	GHS 980	-	-	22

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Accessories & Diagnostics

Tools, safety equipment, and monitoring systems

SKU	Product Name	Price	Original	Save	Stock
ACC-001	EV Dashboard Display 3.5" LCD	GHS 650	GHS 780	17%	40
ACC-002	High Voltage Safety Gloves 1000V	GHS 320	-	-	60
ACC-003	EV Multimeter & Diagnostic Kit	GHS 1,450	GHS 1,700	15%	14

Conversion Kits

Complete retrofit solutions for Ghanaian motorcycle and tricycle fleets

SKU	Product Name	Price	Original	Save	Stock
KIT-001	Motorcycle EV Conversion Kit 72V 3kW	GHS 5,800	GHS 6,500	11%	7
KIT-002	Keke/Tricycle EV Conversion Kit	GHS 12,500	GHS 14,000	11%	5

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Technical Architecture

System Design & Technology Stack

Backend Infrastructure

Runtime	Node.js 18+ with Express.js framework
Database	SQLite (sql.js) with in-memory persistence and file-based backup
Authentication	bcrypt-hashed passwords, session-based auth with role differentiation (customer/admin)
Payments	Paystack API integration for GHS/USD card, mobile money, and bank transfer processing
Email	Nodemailer with Outlook SMTP for order confirmations and quote notifications
Security	CORS, Helmet headers, rate limiting, input sanitization

Frontend Design System

Framework	Vanilla JavaScript SPA (Single Page Application) with custom router
Styling	Custom CSS with CSS variables, no external UI frameworks
Typography	Manrope (Google Fonts) for body, Halfre for display headings
Animations	IntersectionObserver scroll reveals, Ken Burns hero effects, CSS transitions
Responsive	Mobile-first with hamburger navigation, touch-optimized carousels
Icons	Inline SVG stroke icons, zero icon font dependencies

API Endpoints

POST	/api/auth/register	- Customer registration
POST	/api/auth/login	- Session authentication
POST	/api/auth/logout	- Session termination
GET	/api/auth/me	- Current user profile
GET	/api/products	- Product listing with filters
GET	/api/products/:id	- Single product detail
GET	/api/products/categories	- Category metadata
POST	/api/cart/add	- Add item to session cart
PUT	/api/cart/update	- Update cart quantity
DELETE	/api/cart/remove/:id	- Remove cart item
GET	/api/cart	- Retrieve current cart
POST	/api/orders/initialize	- Initiate Paystack payment
GET	/api/orders/my	- Customer order history
POST	/api/quotes	- Submit vehicle sourcing request
GET	/api/blog	- News article listing
GET	/api/blog/:slug	- Single article content
GET	/api/admin/stats	- Dashboard metrics (admin only)

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GET	/api/admin/products	- Product CRUD (admin only)
GET	/api/admin/orders	- Order management (admin only)
GET	/api/admin/quotes	- Quote review (admin only)

User Interface Design

Porsche-Inspired Minimalist System

Design Philosophy

The interface follows a strict monochromatic palette inspired by Porsche's digital presence: pure black, architectural greys, and white space. All rounded corners are removed (border-radius: 0). Typography uses Halfre for headlines - a sharp, condensed display face - and Manrope for body text. The result is a premium, engineering-focused aesthetic that builds trust with technical buyers.

Key Interface Components

Hero Slider

Full-viewport cinematic slides with Ken Burns zoom, staggered text reveals, and 6-second auto-advance. Five slides covering brand, batteries, charging, conversion, and sourcing.

Navigation

Transparent-to-solid transition on scroll. Centered links with animated underlines. SVG icon actions (search, translate, account, cart). Mobile hamburger morphs to X with slide-in drawer.

Product Cards

4:3 aspect ratio images with hover-zoom. Dark minimal badges. 'Add to Cart' slides up on hover (desktop) or always visible (mobile). Clean price row with savings calculation.

Category Cards

16:10 cinematic backgrounds with gradient overlays. Halfre typography for names. Text floats up 3px on hover. No arrows, no gold bars, no decorative clutter.

Cart & Checkout

Side-by-side layout: item list left, sticky summary right. Paystack modal with form validation. Hairline borders, no shadows.

Admin Dashboard

Stat tiles with large numerals. Tabbed panels for products, orders, quotes. Inline status editing. Minimal table design with uppercase headers.

Chatbot

Fixed-position toggle with black/white minimal chat window. Anthropic Claude API integration for EV-specific Q&A.

Footer

Massive 5-column dark footer. Uppercase section headers. Legal bar with social links. Generous 100px vertical padding.

Payments & Logistics

Transaction Flow & Fulfillment

Payment Processing

All transactions route through Paystack, Nigeria's leading payment processor with full Ghanaian GHS support. The checkout flow: customer completes delivery details -> server initializes Paystack transaction -> customer redirected to secure Paystack hosted page -> payment authorization -> webhook confirmation -> order status updated to 'paid' -> confirmation email dispatched.

Supported Payment Methods:

- Visa, Mastercard, Verve cards (local and international)
- Mobile Money (MTN MoMo, Vodafone Cash, AirtelTigo Money)
- Bank transfer and USSD channels

Delivery & Sourcing Logistics

For parts orders: nationwide delivery via local courier partners (estimated 2-5 business days within Accra, 5-10 days to Kumasi/Tamale). Free shipping threshold to be determined based on average order value.

For vehicle sourcing: end-to-end international logistics including supplier verification in China/Europe/North America, export documentation, sea freight to Tema or Takoradi Port, customs clearance with Ghana Revenue Authority, and final inland delivery to customer-specified location.

Quote Workflow

1. Customer submits vehicle specs via quote form (make, model, year, budget, destination port)
2. Admin reviews request and contacts international supplier network for availability and pricing
3. Formal quote prepared with CIF (Cost, Insurance, Freight) or DDP (Delivered Duty Paid) options
4. Customer accepts quote and pays 50% deposit via Paystack
5. Vehicle sourced, inspected, and shipped with full tracking documentation
6. Balance payment due upon arrival at Ghana port or before inland delivery

Marketing & Growth Strategy

Customer Acquisition & Retention

Digital Presence

Instagram	@td_kyeremanteng - Product showcases, EV culture content, customer testimonials
X (Twitter)	@td_kyeremanteng - Industry news, policy updates, technical threads
YouTube	@sakentstudio - Installation tutorials, vehicle sourcing vlogs, battery maintenance guides
WhatsApp	+233 59 477 5210 - Primary sales and support channel for Ghanaian market

Content & SEO Strategy

- Blog publishing 2x weekly on EV policy, battery technology, and Ghana-specific adoption challenges
- Google Business Profile optimization for 'EV parts Accra' and 'electric vehicle conversion Ghana'
- Technical documentation and datasheets for every product (builds B2B trust)
- Partnership with Ghanaian mechanic associations and polytechnic engineering departments
- Referral program for conversion kit installers (commission on referred parts sales)

Customer Retention Mechanics

Order Tracking

Real-time SMS and email updates from processing through delivery

Loyalty Program

Points system for repeat purchases, redeemable on accessories and tools

Technical Support

Lifetime WhatsApp access for installation guidance and troubleshooting

Warranty Registration

Online warranty activation with serial number tracking

Newsletter

Monthly EV market report with exclusive subscriber pricing

Admin Operations

Back-Office Management System

Dashboard Metrics

The admin panel provides real-time visibility into store performance through five key stat tiles: total orders, cumulative revenue (GHS), active product count, registered users, and pending quote requests. All metrics update automatically via API calls to the SQLite database.

Product Management

Full CRUD interface for the 16-SKU catalog. Admin can add new products with automatic category assignment, edit pricing and stock levels inline, set featured flags for homepage promotion, and delete discontinued items. All changes persist to the SQLite database immediately.

Admin Credentials (Demo):

Email: admin@evshopgh.com

Password: evadmin2024

Order Fulfillment Pipeline

PENDING	Order received, payment pending or processing
PROCESSING	Payment confirmed, picking and packing initiated
SHIPPED	Handed to courier, tracking number assigned
DELIVERED	Confirmed receipt by customer, warranty period begins

Quote Management

Vehicle sourcing quotes follow a four-stage pipeline: NEW (unreviewed), REVIEWING (admin assessing supplier options), CONTACTED (quote sent to customer), CLOSED (order placed or declined). Status changes trigger optional email notifications to the customer.

Development Roadmap

Current State & Future Releases

Version 1.0 - Current (May 2026)

- Complete 16-SKU product catalog with category filtering and search
- Paystack payment integration for GHS transactions
- Session-based shopping cart with quantity management
- Customer authentication and order history
- Admin dashboard with product/order/quote CRUD
- Vehicle sourcing quote submission system
- Responsive SPA frontend with Porsche-inspired minimal design
- AI chatbot integration (Claude API) for EV Q&A
- WhatsApp direct sales channel
- Google Translate widget for multilingual support

Version 1.1 - Q3 2026 (Planned)

- PostgreSQL migration for production-scale persistence
- Cloudinary image CDN integration with automatic WebP conversion
- Inventory management with low-stock alerts
- Bulk product import via CSV for supplier catalog updates
- Customer review and rating system
- Abandoned cart recovery email sequence
- Advanced analytics dashboard (conversion rates, cohort retention)

Version 2.0 - 2027 (Strategic)

- Mobile app (React Native) for field technicians and fleet managers
- B2B wholesale portal with tiered pricing and credit terms
- Integration with Ghana Grid Company for charging station mapping
- EV financing partnerships with Stanbic, Ecobank, and CalBank
- Battery recycling and trade-in program
- Franchise model for regional distribution centers (Kumasi, Tamale, Takoradi)

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Contact & Appendix

Project Metadata

Key Contacts

Business Email	evsolutionsgh@outlook.com
WhatsApp Sales	+233 59 477 5210
Website	evshopgh.com
Instagram	@td_kyeremanteng
X / Twitter	@td_kyeremanteng
YouTube	@sakentstudio

File Structure Reference

```
ev-shop-gh/
- server.js           # Express server, security, SPA routing
- database.js         # SQLite schema, seed data
- mailer.js           # Email templates (Outlook SMTP)
- package.json        # Dependencies
- .env                # Environment variables
- routes/
  - auth.js           # Login, register, logout, me
  - products.js       # Product CRUD, categories, search
  - cart.js           # Session cart management
  - orders.js         # Paystack checkout, order history
  - quotes.js         # Vehicle sourcing requests
  - blog.js           # News articles
  - admin.js          # Admin-only CRUD endpoints
  - currency.js       # IP-based currency detection
- public/
  - index.html        # Complete SPA frontend
  - fonts/            # Halfre typeface files
  - images/           # Product & category photos
    - products/       # 16 SKU images (local hosting)
    - categories/     # 6 category backgrounds
    - hero/           # 5 hero slider images
- evshop.db           # SQLite database file
```

Technology Versions

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Node.js	18.x LTS or higher
Express	4.18.x
SQLite (sql.js)	3.46.x
Paystack API	2023-10 (current)
Nodemailer	6.9.x
bcryptjs	2.4.x

Security Considerations

- PAYSTACK_SECRET_KEY must be rotated from test to live before production deployment
- EMAIL_PASS should use Outlook App Password (not account password) if 2FA is enabled
- SQLite database file must be on persistent disk for cloud deployments (Render, Railway)
- Admin credentials should be changed immediately after first login
- CORS origin should be restricted to evshopgh.com in production
- Rate limiting recommended on /api/orders/initialize to prevent payment spam

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